

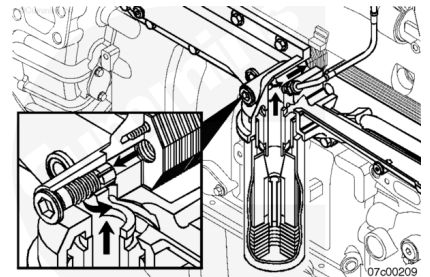
Trainings ▾

General Information

Whenever the pressure drop across the lubricating oil filter exceeds 862 kPa [125 psi], the oil filter bypass valve opens and allows lubricating oil to bypass the lubricating oil filter.

This condition can occur during cold ambient (cold lubricating oil) engine start-ups.

The purpose of the bypass valve is to maintain lubricating oil flow to the engine and prevent an oil filter collapse.



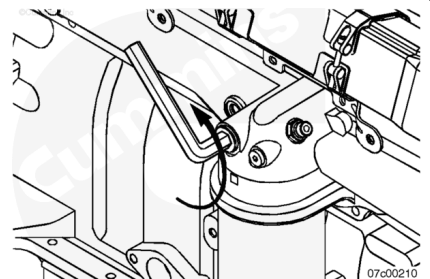
Remove

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

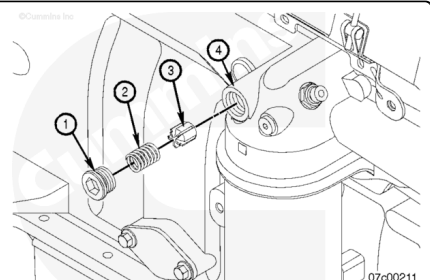
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



Loosen the retaining plug with a 7-millimeter Allen wrench.

Remove the retaining plug (1), spring (2), and plunger (3) from the lubricating oil cooler housing (4).



Clean

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

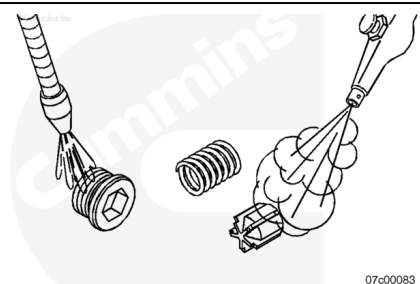
⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

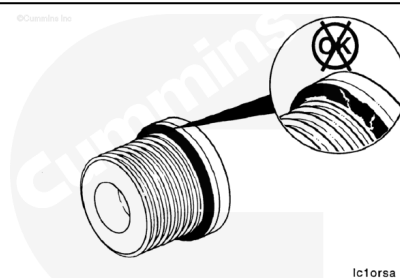
Clean the spring, plunger, and retaining plug with solvent. Dry with compressed air.



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Inspect for Reuse

Inspect the o-ring on the retaining plug for cuts or tears. Replace the o-ring if damaged.



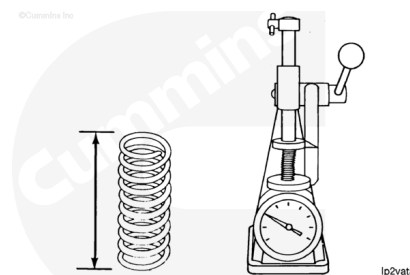
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Use a spring compression tester, Part Number 3375182, to measure the relief valve spring tension.

Compress the spring to 29.65-mm [1.17-in] height. The force required to compress the spring **must** be:

n		lbf
115.7	NOM	26

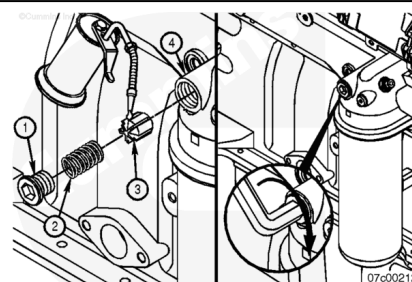
If the spring does **not** meet this specification, it **must** be replaced.



Install

⚠ CAUTION ⚠

The notched end of the bypass valve (3) must be installed toward the spring (2). If the bypass valve is installed incorrectly, it will bypass the filter continuously which can cause engine damage.



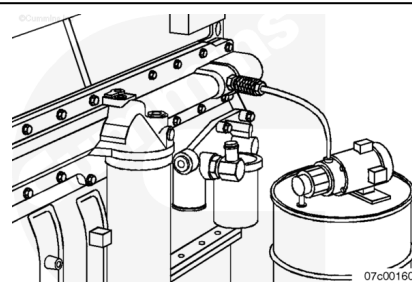
Lubricate all pieces and the o-ring seal. Install the plunger (3) with the notched end toward the spring into the lubricating oil cooler housing (4).

Install the spring (2), and retaining plug (1) into the lubricating oil cooler housing (4).

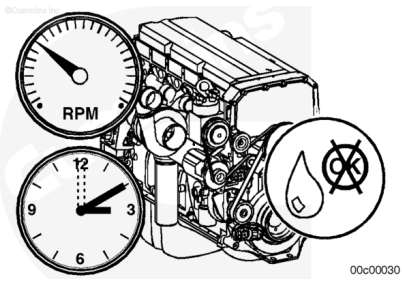
Tighten the retaining plug.

Torque Value: 45 n•m [33 ft-lb]

Prime the lubricating oil system. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/10/10-007-037.html>).



Operate the engine to normal operating temperature and check for leaks.



Last Modified: 26-Jun-2002
